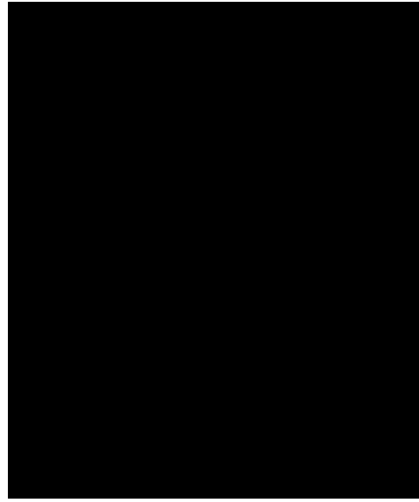


Mühendislik Fakültesi
Bilgisayar Mühendisliği Bölümü
Sayısal Gözümleme Arasınav Soruları
Bahar 2019-2020 Uzakta Eğitim



[REDACTED]

$$\ln(x-1) + \cos(x-1) = 0$$

$\Lambda = 1$

$n=2$

$$|1,397320 - 1,391847| = 0,005473 > 10^{-5}$$

$n=3$

$$|1,397741 - 1,397320| = 0,00042 > 10^{-5}$$

$n = 4$

$$|1,3977485 - 1,3977811| = 4 \times 10^{-7} < 10^{-5}$$

Soru 2)

x	0,2	0,4	0,6	0,8	1,0
f(x)	0,9798652	0,917710	0,8080380	0,6386093	0,3843735

a) $f'(0,4)$ ve $f''(0,4)$ türevlerini yaklaşık hesaplayan tüm mümkün durumlar

$f'(0,4)$ için 1. adım $\Rightarrow$ 3 nokta ve nokta

$$h=0,2$$

$$f'(x_0) = \frac{1}{2h} [-3f(x_0) + 4f(x_0+h) - f(x_0+2h)]$$

$$f'(0,4) = \frac{1}{0,4} [-3f(0,4) + 4f(0,2) - f(0)] = \underline{-0,399425}$$

2. adım $\Rightarrow$ 3 nokta ve nokta $h=-0,2$

$$f'(x_0) = \frac{1}{2h} [-3f(0,4) + 4f(0,2) - f(0)] = \text{hesaplamak gerekmiyor}$$

$f(0)$ değerini bilmiyoruz.

3. adım $\Rightarrow$ 3 nokta orta nokta

$$f'(x_0) = \frac{1}{2h} [f(x_0+h) - f(x_0-h)] \quad h=0,2$$

$$f'(0,4) = \frac{1}{0,4} [f(0,6) - f(0,2)] = \underline{0,423568}$$

• $f''(0,4)$ için $\Rightarrow$ 1. yöntem orta nokta

$$f''(x_0) = \frac{1}{h^2} [f(x_0-h) - 2f(x_0) + f(x_0+h)] \quad h=0,2$$

$$f''(0,4) = \frac{1}{0,04} [f(0,2) - 2f(0,4) + f(0,6)] = \underline{-1,9097}$$

soru 2)

b) $f'(0,6)$ ve $f''(0,6)$ yaklaşık hesaplayın tüm mümkün durumlar değerini bul

* $f'(0,6)$ için $\rightarrow$ 1. adım 3 nokta uç nokta $h=0,2$

$$f'(0,6) = \frac{1}{0,4} [-3f(0,6) + 4f(0,8) - f(1,0)] = \underline{0,635125}$$

$\rightarrow$ 2. adım 3 nokta orta nokta $h=0,2$

$$f'(0,6) = \frac{1}{0,4} [f(0,8) - f(0,4)] = \underline{-0,697904}$$

$\rightarrow$ 3. adım 3 nokta orta nokta $h=0,2$

$$f'(0,6) = \frac{1}{0,4} [f(0,8) - f(0,4)] = \underline{0,697904}$$

$\rightarrow$ 4. adım 5 nokta orta nokta $h=0,2$

$$f'(x_0) = \frac{1}{12h} [f(x_0-2h) - 8f(x_0-h) + 8f(x_0+h) - f(x_0+2h)]$$

$$f'(0,6) = \frac{1}{2,4} [f(0,2) - 8f(0,4) + 8f(0,8) - f(1,0)] = \underline{-0,682308}$$

* $f''(0,6)$ için 1. adım
 $h=0,2$ için orta nokta

$$f''(0,6) = \frac{1}{(0,2)^2} [f(0,4) - 2f(0,6) + f(0,8)] = \underline{-1,492392}$$

$\rightarrow h=0,4$ için orta nokta

$$f''(0,6) = \frac{1}{(0,4)^2} [f(0,2) - 2f(0,6) + f(1,0)] = \underline{-1,573983}$$

örv 3) $f(x) = 2x + 3\cos x - e^x$
 $f(x_n) = 2x_n + 3\cos x_n - e^{x_n}$

$$\begin{matrix} x_0 = 1 \\ x_1 = 2 \end{matrix}$$

$$f(x_0) = 0,09026250$$

$$f(x_1) = -4,63749660$$

$$x_n = x_{n-1} - \frac{(x_{n-1} - x_{n-2}) \cdot (2x_{n-1} + 3\cos x_{n-1} - e^{x_{n-1}})}{(2x_{n-1} + 3\cos x_{n-1} - e^{x_{n-1}}) - (2x_{n-2} + 3\cos x_{n-2} - e^{x_{n-2}})}$$

n=2 için

$$x_2 = x_1 - \frac{(x_1 - x_0) \cdot (2x_1 + 3\cos x_1 - e^{x_1})}{(2x_1 + 3\cos x_1 - e^{x_1}) - (2x_0 + 3\cos x_0 - e^{x_0})}$$

$$x_2 = 2 - \frac{1 \cdot (4 - 1,2448440 - 7,389056)}{4 - 1,2448440 - 7,389056 - 2 - 1,620906 + 2,718281}$$

$$x_2 = 1,162925$$

$$|x_2 - x_1| < 10^{-5} \text{ değil}$$

$$|1,162925 - 2| = 0,837075 > 10^{-5} \text{ (DEVAM)}$$

n=3 için

$$x_3 = x_2 - \frac{(x_2 - x_1) \cdot (2x_2 + 3\cos x_2 - e^{x_2})}{(2x_2 + 3\cos x_2 - e^{x_2}) - (2x_1 + 3\cos x_1 - e^{x_1})}$$

$$x_3 = 1,162925 - \frac{-0,264368}{4,95408}$$

$$x_3 = 1,216410$$

$$|x_3 - x_2| = 0,053485 > 10^{-5} \text{ (DEVAM)}$$

Sayfa ①

Soru 3)

$n=4$ için

$$x_4 = x_3 - \left(\frac{(x_3 - x_2) \cdot (2x_3 + 3\cos x_3 - e^{x_3})}{2x_3 + 3\cos x_3 - e^{x_3} - 2x_2 - 3\cos x_2 + e^{x_2}} \right)$$

$$x_4 = 1,216410 - \left(\frac{0,005285}{-0,217725} \right)$$

$$\boxed{x_4 = 1,24068}$$

$$|x_4 - x_3| = 0,02427 > 10^{-5} \text{ (DEVAM)}$$

$n=5$ için

$$x_5 = x_4 - \left(\frac{(x_4 - x_3) \cdot (2x_4 + 3\cos x_4 - e^{x_4})}{2x_4 + 3\cos x_4 - e^{x_4} - 2x_3 - 3\cos x_3 + e^{x_3}} \right)$$

$$x_5 = 1,24068 - \left(\frac{-0,000100}{-0,102959} \right)$$

$$\boxed{x_5 = 1,23970}$$

$$|x_5 - x_4| = 0,00087 > 10^{-5}$$

$n=6$ için

$$x_6 = x_5 - \left(\frac{(x_5 - x_4) \cdot (2x_5 + 3\cos x_5 - e^{x_5})}{2x_5 + 3\cos x_5 - e^{x_5} - 2x_4 - 3\cos x_4 + e^{x_4}} \right)$$

$$x_6 = 1,23970 - \left(\frac{-6,181726 \cdot 10^{-3}}{0,062079} \right)$$

$$\boxed{x_6 = 1,23969}$$

$$|x_6 - x_5| = 0,0000013 < 10^{-5}$$

$$\boxed{\text{Kök} \Rightarrow x_6 = 1,23969}$$

sayfa ②

soru 4)

$$\int_0^2 x^2 \ln(x^2+1) dx$$

$$h=0,25$$

a) Birleştirilmiş yamuk

$$0,25 = \frac{2}{n}$$

n=8

$$I = \frac{1}{8} [f(0) + 2f(0,25) + 2f(0,5) + 2f(0,75) + 2f(1) + 2f(1,25) + 2f(1,5) + 2f(1,75) + f(2)]$$

$$I = \frac{1}{8} \cdot [25,27577]$$

$$I \approx 3,15947$$

$$f(0) = 0$$

$$2 \cdot f(0,25) = 0,00757$$

$$2 \cdot f(0,5) = 0,11157$$

$$2 \cdot f(0,75) = 0,50207$$

$$2 \cdot f(1) = 1,38629$$

$$2 \cdot f(1,25) = 2,94057$$

$$2 \cdot f(1,5) = 5,30394$$

$$2 \cdot f(1,75) = 8,58601$$

$$f(2) = 6,43775$$

b) Birleştirilmiş Simpson

$$0,25 = \frac{2}{n}$$

n=8

$$I = \frac{1}{12} [f(0) + 4f(0,25) + 2f(0,5) + 4f(0,75) + 2f(1) + 4f(1,25) + 2f(1,5) + 4f(1,75) + f(2)]$$

$$I = \frac{1}{12} \cdot [37,31223]$$

$$I \approx 3,10935$$

$$f(0) = 0$$

$$4 \cdot f(0,25) = 0,01512$$

$$2 \cdot f(0,5) = 0,11156$$

$$4 \cdot f(0,75) = 1,00914$$

$$2 \cdot f(1) = 1,38628$$

$$4 \cdot f(1,25) = 5,88114$$

$$2 \cdot f(1,5) = 5,30394$$

$$4 \cdot f(1,75) = 17,1723$$

$$f(2) = 6,43775$$

sayfa ①

$$c) \quad 0,25 = \frac{2}{n}$$

$$n=8$$

$$\int_0^2 x^2 \ln(x^2+1) dx$$

$$\int_a^b f(x) dx = 2h \sum_{j=0}^{n/2} f(x_{2j})$$

$$I = 0.5 [f(0,25) + f(0,75) + f(1,25) + f(1,75)]$$

$$I = 0.5 [6,018119]$$

$$I \approx 3,00906$$

$$f(0,25) = 0,003789$$

$$f(0,75) = 0,251036$$

$$f(1,25) = 1,470286$$

$$f(1,75) = 4,293008$$

Soru 5)

$$a) \int_0^2 e^{2x} \sin 3x dx$$

$$f(x) = e^{2x} \sin 3x$$

$$f'(x) = 3e^{2x} \cos 3x + 2e^{2x} \sin 3x$$
$$= e^{2x} (3 \cos 3x + 2 \sin 3x)$$

$$f''(x) = -5e^{2x} \sin 3x - 12e^{2x} \cos 3x$$
$$= e^{2x} (12 \cos 3x - 5 \sin 3x)$$

$$\left| \frac{h^3}{12n^2} f''(\xi(x)) \right| \Rightarrow f'(0) = 12$$

$$f'(2) = 705,360102876$$

$$f''(2) \leq 705,36$$

$$|I_1| \leq \frac{2^3}{12n^2} \cdot 705,36 < 10^{-4}$$

$$|I_2| \leq \frac{2}{3n^2} \cdot 705,36 \leq 10^{-4}$$

$$\sqrt{n^2} \geq \sqrt{\frac{2 \cdot 705,36}{3 \cdot 10^{-4}}} \Rightarrow n \geq \sqrt{\frac{2 \cdot 705,36}{3 \cdot 10^{-4}}}$$

$$n \geq 2168,501$$

$$h = \frac{b-a}{n} = \frac{2}{2168,501}$$

$$h = \frac{2}{2168,501} = 0,000922295$$

Soru 5)

$$b) \frac{b-a}{180} \cdot h^4 \cdot f^{(4)}(\xi(x)) \leq 10^{-4}$$

$$\begin{aligned} f'''(x) &= 9e^{2x} \cos 3x - 46e^{2x} \sin 3x \\ &= e^{2x}(9\cos 3x - 46\sin 3x) \\ f'''(x) &= -119e^{2x} \sin 3x - 120e^{2x} \cos 3x \\ &= -e^{2x}(120\cos 3x + 119\sin 3x) \end{aligned}$$

$$f^{(4)}(x) = -e^{2x}(119\sin(3x) + 120\cos(3x))$$

$$\frac{2}{180} \cdot h^4 \cdot f^{(4)}(2) \leq 10^{-4}$$

$$h^4 \leq \frac{90 \cdot 10^{-4}}{4475.4098} \Rightarrow h \leq 0.037658$$

$$h = \frac{b-a}{n} \Rightarrow n = \frac{b-a}{h}$$

$$n \geq 53.11 \Rightarrow n \geq 53$$

$$h < 0.037658$$

$$c) \frac{b-a}{6} h^2 f''(\xi(x)) \leq 10^{-4}$$

$$f^{(2)}(x) = e^{2x}(12\cos(3x) - 5\sin(3x))$$

$$\frac{2}{6} \cdot h^2 \cdot (705.360103) \leq 10^{-4}$$

$$f^{(2)}(2) = 705.360103$$

$$h^2 \leq \frac{3 \cdot 10^{-4}}{705.360103}$$

$$h \leq (6.521615) \cdot 10^{-4}$$

$$n = \frac{2}{h} - 2$$

$$n \leq 3066.72$$

$$n \leq 3066.72$$

$$h < 0.00065216$$

$$n \geq 3066$$

soru 6)

$$f(x) = e^{3x} - 4x$$

$$f'(x) = 3e^{3x} - 4$$

$$f''(x) = 9e^{3x}$$

$$f'''(x) = 27e^{3x}$$

$$h_0 = \sqrt[3]{\frac{3\varepsilon}{M_3}}$$

$$\varepsilon = 0,5 \times 10^{-4}$$

$$M_3 = \max_{[0,3,0,7]} |f'''(x)|$$

$$M_3 = \max |27e^{3x}| = |27e^{3 \cdot 0,7}|$$

$$M_3 \approx 220,486587$$

$$h_0 = \sqrt[3]{\frac{3 \cdot 0,5 \cdot 10^{-4}}{220,486587}}$$

$$h_0 \approx 0,008795$$